

# CSE 370 Database Systems

## Assignment 1

### Section 22

### Spring 25

**1.** At Hogwarts School of Witchcraft and Wizardry, Quidditch is more than just a sport—it's a way of life. The Hogwarts Quidditch Club manages everything from house teams to tournaments, ensuring that the game runs smoothly. The club keeps track of players, advisors, house teams, Quidditch matches, match results, and other magical events.

As a database designer, you have been tasked with building an Enhanced Entity-Relationship (EER) diagram for the club. Your design must capture the relationships between *players*, *teams*, *matches*, and *their results* while following these magical constraints: **[10]**

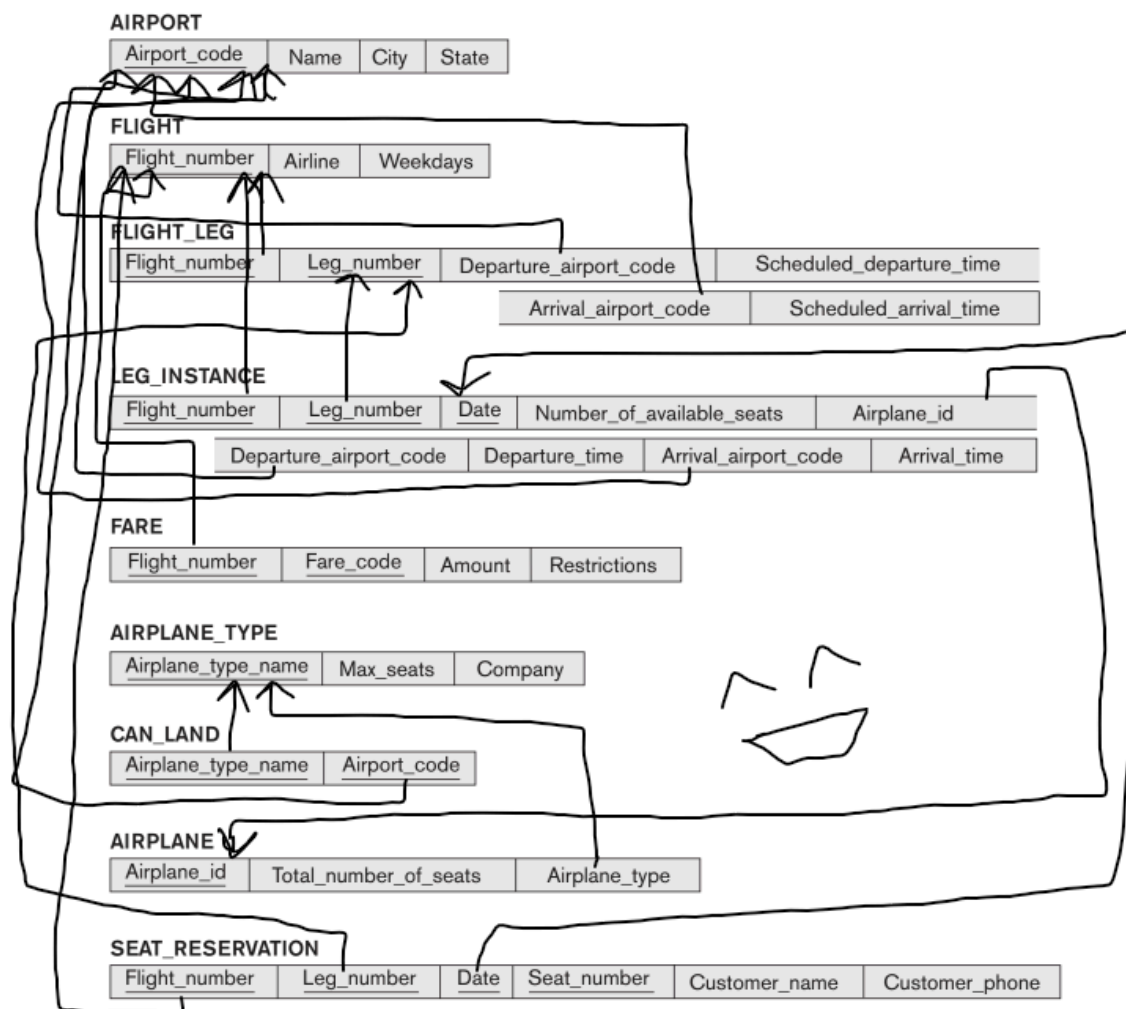
1. Every player at Hogwarts must belong to a house team. However, not all players perform the same role on the field. Some take on specialized roles such as Chasers, Beaters, Keepers, or Seekers. Since every player must fit into exactly one of these roles, your design should clearly distinguish between them. Additionally, make sure each specialized role has unique attributes that set them apart (e.g., Chasers focus on scoring goals, Keepers on blocking shots, etc. [You can make assumptions here or just mention these two.] ).
2. The database must include at least four entities (excluding subclasses).
3. Some experienced players mentor younger players to help them improve their skills so you must represent this in your diagram.
4. Players do not just play for their teams—they also participate in various Quidditch tournaments, where different house teams compete. Since multiple teams can compete in multiple tournaments keep that in mind.
5. Each Quidditch match played at Hogwarts has a scoreboard, which records details such as total points scored, house team scores, and time remaining. However, a scoreboard is only meaningful when associated with a specific match. If a match is removed from the records, the corresponding scoreboard will also lose its significance.
6. Players train rigorously, and their total training hours should be recorded. However, rather than storing this as a fixed value, it must be determined dynamically based on the individual *practice sessions* a player has attended.

Your EER diagram should depict important attributes of all entities and relationships, ensuring that the Hogwarts Quidditch Club's magical database is well-structured and ready for the next big match!

**2.** Consider the AIRLINE database schema that manages information about flights, flight legs, seat reservations, airplanes, and airports. The database enforces several integrity constraints, such as primary keys, foreign keys, not-null constraints, and referential integrity.

For each of the following operations, discuss whether it can violate any integrity constraints or, not **[As the Database is not given you are allowed to make assumptions]**. If an operation violates constraints, explain the issue in detail.

[5]



1. Insert <'DL101', '1', 'JFK', '2025-08-01 10:00:00', 'LAX', '2025-08-01 14:00:00'> into FLIGHT\_LEG.
2. Insert <'LAX', 'Delta Lounge', 100, 'CA'> into AIRPORT.
3. Delete the FLIGHT\_LEG tuple where Flight\_number = 'BA215' and Leg\_number = '2'.
4. Delete the FLIGHT tuple where Flight\_number = 'EK205'.
5. Update SEAT\_RESERVATION where Customer\_phone = '1234567890' to set Seat\_number = 'A12'